

Attention orienting and serial search in Virtual Reality

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Abstract

Orienting attention in space is crucial for efficient perception and learning. The cognitive and neural underpinnings of spatial attention have been extensively studied in tasks requiring focusing and shifting attention across locations. However, these studies mainly explored frontal 2D visual space. Here we designed a virtual reality (VR) environment combining a visual search task with a modified Posner orienting paradigm to investigate the deployment of attention in both front and rear locations, and compare performance with classic 2D findings. We combined response times with eye-tracking data recorded in the VR headset. Twenty participants were asked to discriminate between 2 targets arranged in 4 locations around them. Targets were spatially cued (20% Valid, 80% Invalid), and presented with a varying number of identical distractors (Zero, One Near, One Far, Three) to additionally probe serial search. Target detection was faster in valid than invalid conditions, but delayed by distractors, in line with classic effects in Posner and search tasks, respectively. Eye-tracking data confirmed that attention shifts, as observed by gaze patterns over time, were tightly correlated with response times, and both were modulated by cue validity and distractor load. Our results confirm eye-tracking is a powerful tool in VR for quantifying attentional dynamics through overt gaze strategies. The behavioral and gaze data converge for targets appearing in rear or front space, suggesting that attentional orienting engages similar mechanisms of sensory competition and selection in a full 360° space (in both front and rear), supporting a functional continuity in space representation used to guide selective attention processes.

Keywords: Virtual Reality, Spatial Attention, Eye-tracking, Auditive Cueing, Visual Attention, Non-Visual Attention

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INTRODUCTION

Abundant research in psychology and cognitive neuroscience has studied attention and how it allows selective perception. Distinct mechanisms control attention, including exogenous bottom-up components triggered by salient stimuli, such as a sudden loud noise in a quiet room or a flashing light in the peripheral field of vision, and endogenous top-down components driven by goals, memory, or motivation. The contribution of these different factors has been explored with multiple paradigms, for instance, by looking for a specific target in a crowded place (Corbetta & Shulman, 2002; Vuilleumier, 2015). These attentional mechanisms amplify sensory responses, modulate stimulus awareness, and guide behaviour. A crucial aspect of selective attention is that it operates dynamically through successive engagement at different locations in space, metaphorically described as a “spotlight” effect (Gaillard & Ben Hamed, 2022). While attention is rapidly attracted to salient stimuli that “pop out”, it can be directed to locations and objects based on task goals (Wolfe, 2021) - generating a slower serial search.

Reorienting attention to new locations tends to slow performance, reflecting the cost of attention shifts across space. This effect is classically demonstrated in the Posner paradigm (Chica et al., 2013; Posner, 1980), where participants respond faster when a visual or auditory cue correctly predicts where a target will appear, but become slower when the cue indicates a false position. This is classically attributed to the participants having to disengage from a given location to reorient their attention towards the relevant location (Prime et al., 2006; Rizzolatti et al., 1987).

Although this is a widely replicated and accepted interpretation, most research has been conducted using 2D screens, simplified stimuli, and a relatively small area for visual exploration. These tasks differ substantially from real-life situations where searching for an object in our surroundings requires simultaneous eye, head and body movements. Only a few studies (Gilchrist et al., 2001; Golomb, 2019; Smith et al., 2008) examined differences in search behaviour between naturalistic 3D (layout in rooms) and 2D environments (tasks on screens). Their findings suggest that visual search efficiency may vary depending on spatial scale, the presence of visual guidance, and the physical demands of exploration. Specifically, when targets are visually distinctive (feature-present or feature-absent), search profiles remain consistent across 2D and large-scale 3D spaces. However, when participants must physically navigate and inspect locations without visual cues, closer to natural foraging, search becomes markedly slower and engagement of spatial memory increases substantially, with far fewer revisit errors than predicted by traditional eye-movement studies.

Other studies concentrating on navigation and scene viewpoint agree that a more complex setting (e.g. various objects, movement, colour, etc.) leads to higher exploration demands across the visual space (Franchak, 2020; Wu & Tsotsos, 2025). The higher complexity, the more time is necessary to treat sensory information, although participants usually get better in terms of efficiency over time. This suggests learning effects whereby some hierarchy is installed to decipher important and less important information in the environment.

Effective visual search in the real world also implies larger and more complex cognitive representations of 3D space. Such maps might differ between objects located in the visible front space, which rely mainly on visual perception, and objects situated in the rear space, which depend more on memory. Generally, there has been little exploration on how attention

deploys in 3D space, and whether classic paradigms extend to both front and rear space. Very few studies have examined attention and search in 3D, either in the real-world (Foulsham et al., 2014; Maya et al., 2019) or in VR environments (De Lillo & James, 2012; Ruddle & Lessels, 2006), and most of them examined search in 3D scenes seen in front of the observer but did not consider rear space. Some studies compared attention in front and rear using sounds behind participants or projecting visual stimuli on mirrors (Soret et al., 2022). They showed that cross-modal interactions occur in 3D layouts and spatial auditory cues attract attention toward visual stimuli in rear space, but mirror experiments do not generalize to real-life situations beyond car driving (Ho & Spence, 2005). Recent work (Soret et al., 2022) used front and rear stimuli shown on virtual screens in 3D VR scenarios (e.g. spaceship cockpit), and confirmed that spatial cueing effects occur in both front and rear locations. However, they also reported slower attention shifts in the rear, suggesting a different distribution of attention for events behind the observer. Other work exploited the 3D immersive capacity of VR (Harada & Ohyama, 2019, 2020) with very simple environments to characterize detection performance as well as eye and head movement patterns in visual search. The latter concluded that attention deployment in 3D is influenced by task-irrelevant spatial context, such as a plane layout (no plane, basic plane and room in virtual reality). The more details the environment has, even irrelevant to the task, the more the space is searched and explored (Harada & Ohyama, 2020). Such strategies to detect three-dimensional structures affect attentional allocations thus explaining the difference in performances in different layouts.

In the present study, we designed a novel VR-based paradigm combining search with a Posner task to examine how attention is oriented in both front and rear space. Our primary aim was to assess whether the fundamental principles of spatial attention established in classical laboratory tasks extend to immersive VR environments and, more precisely, compare attention orienting between front and rear space in order to determine the inherent characteristics of attention in the latter, and examine whether they can be compared to the known cognitive processes in front space. Serial search effects and spatial shifts of attention suggest that attentional processes operate across both front and rear space in a comparable manner. Eye-tracking data provides complementary evidence of such attentional shifts, consistent with behavioural performance.

PROCEDURE AND METHODOLOGY

1. Participants

Twenty-five participants took part in the study ($M = 24.04$ years, $SD = 3.30$) including 17 women and 5 left-handed participants. All had normal to corrected vision and normal hearing. The experiment was approved by the Cantonal Research Ethics Committee (CCER) of Geneva. Written informed consent was obtained from all participants before the experiment. All participants received a compensation of 20 CHF after their participation.

Five participants were excluded due to a high rate of incorrect responses (no correct response for at least one condition). One more participant was excluded from the eye-tracking analysis due to technical issues with the recording of eye movements.

2. Apparatus

We used the HTC Vive Focus Vision VR headset with an integrated eye-tracker and HTC Vive controllers to present the stimulus and to collect the user responses, the user movements (head, hand controllers) and the eye-tracking data. The VR headset has a 120-degrees horizontal field of view and 110-degree vertical field of view. We used external headphones, the DT 770 M from Beyerdynamic, to improve the quality of sound spatialization. The VR environment was realized on Unity3D 6000.0.34f1. 3D Models were obtained from the Unity3D asset store. Participant made responses using two buttons on the VR wand. They could use either a left or right held wand, depending on their dominant hand. To report targets, participants had to press one of two buttons on the controllers, depending on the target. Each of these buttons was assigned to a different object target, counterbalanced between participants.

3. Stimuli and Task

The VR environment depicted a room full of toys. Two toys were used as targets and a third toy was presented on one or several locations to create distractors. Both targets and distractors were presented in front/visible and rear/non-visible space to induce shifts of attention in naturalistic 3D space conditions. The targets and distractors were purposefully chosen to share features (round shape, same colour) to increase difficulty and impose focused attention during the discrimination task.

The experiment followed a Posner-like design combined with visual search. The task was to identify one of two target toys (pink teddy bear or pink unicorn). The target appeared at a different position in each trial and was shown for a maximum of 3 seconds. Participants had to report which target was presented (e.g. bear or unicorn), by pressing one of two buttons on the VR wand, as fast as possible. The Posner effect was induced by an auditory spatialized cue preceding target onsets. We chose spatialised auditory cues, unlike visual cues used in classic Posner paradigm, in order to orient attention to either front or rear space in a similar manner. The cue was taken from the Freesound online database.

Serial search was induced by presenting a distractor toy (a pink cat) in one or several locations. Targets and distractors appeared on one of four pillars presented around the participant at $+45^\circ$ and -45° on the front side and at $+135^\circ$ and -135° on the rear side. Thus, there were four task-relevant positions: front left, front right, rear left, and rear right.

Distractors were presented in four configurations (see Figure 1): no distractor; one distractor near, i.e., same spatial sector (front or rear) but opposite side of the target (90° away); one distractor far, i.e., opposite side and opposite sector from position of the target (180° away); or three distractors, i.e., one on each of the three other pillars where the target did not appear. This factor allowed introducing a search component during target detection, allowing us to probe for effects of competition and serial shifts in orienting attention.

The order of targets, as well as the number and position of distractors, were randomized across participants.

In the VR environment, a TV screen was also visible, placed 153 cm above the ground and 170 cm in front of the participant's position. A fixation cross appeared on this screen at the

beginning of each trial. After 3 seconds of maintained fixation, a spatial auditory cue (200Hz tone) was presented at one of the four pillar positions (right/left x front/rear), followed by the target toy (SOA = 150 ms) (Buchtel & Butter, 1988). Each trial lasted a maximum of 3 seconds, followed by a 4 seconds inter-trial interval. This interval allowed the participant to get back in place in front of the TV screen and fixate the cross presented on it - before the start of the next trial.

4. Experimental design

Cue validity was manipulated within-subject by varying whether the auditory cue accurately predicted the upcoming target location (valid, 80% of trials) or not (invalid, 20% of trials), similar to classic attention paradigms (Posner, 1980; Soret et al., 2024). The invalid cue was always presented contralateral to the position of the target, but in the same (front or rear) sector of space. All conditions were randomised and equally likely within each block (see Table 1).

Response times and accuracy were recorded for all participants across all trials, resulting in a $2 \times 2 \times 4$ factorial design with three within-subject factors—Spatial sector (Front vs. Rear), Cue validity (Valid vs. Invalid), and Distractor condition (Zero, One near, One far, and Three) (table 1).

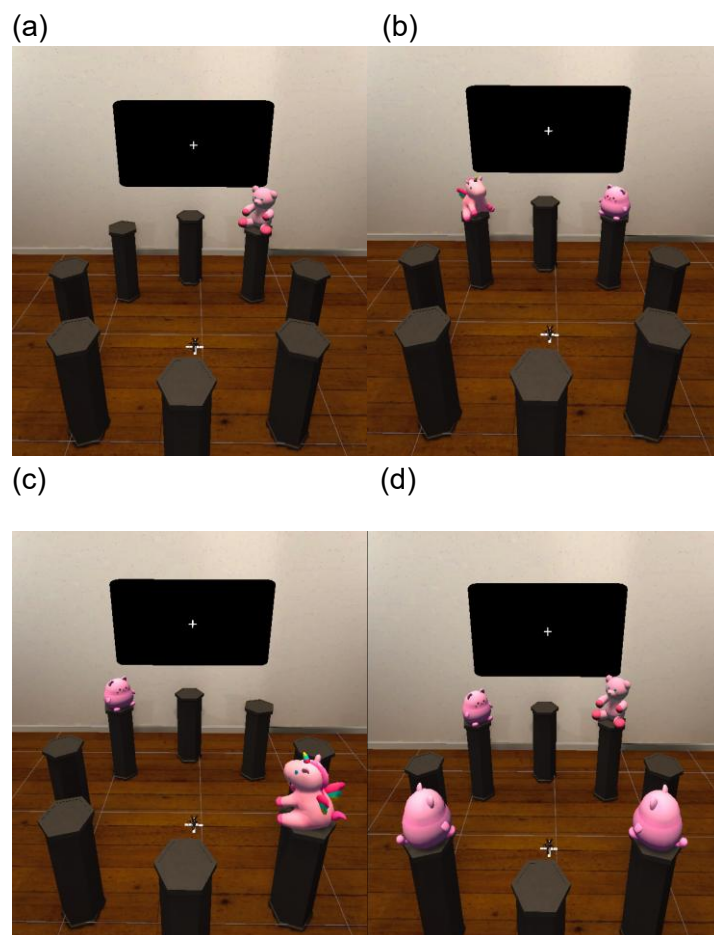


Figure 1: Toy room with 8 pillars arranged around the participant. Only 4 of these pillars were

used to present targets. The cross at the center of pillars indicates the position of the participant at the start of each trial. (a) Example of one of the two possible target objects, the teddy bear, here with no distractor. (b) Example of the other possible target, the unicorn, with one distractor, the toy cat, here at a near location (rotation of 90 degrees from the target). (c) Example of one target with one distractor at a far location (180 degrees from the target). (d) Example of one target with three distractors.

5. Procedure

All participants signed an informed consent form and answered a questionnaire concerning their age, gender, dominant hand, and the language they were most fluent in (English or French). Then they put on the VR headset. Once exposed to the VR scene, they were asked to choose an avatar among 2 options to match their gender.

Instructions were explained orally by the experimenter and then again presented in VR. After answering any question the participants may have had, the experimenter placed the headphones on the headset. Participants had to face the TV at the beginning of each trial and were asked to not move their feet in order to preserve the front and rear spatial sectors identical during the trial.

Participants first went through a training session, composed of 10 trials. They subsequently performed 4 experimental blocks, each composed of 40 trials, in which all conditions were presented and randomised. During the experiment, participants had three 2-minute breaks, during which they took off the headset to prevent any motion sickness induced by movements in the VR environment. After breaks, participants had to press on one of the keys on the handles to continue the experiment. The session lasted a total of 40 minutes, with 20 minutes for the VR task.

6. Data processing and analysis

6.1. Filtering the data

For a total of 3200 trials (160 trials per participant), we first excluded trials with a response time under 100 ms, considered too fast for human choice performance (Harada & Ohyama, 2020; Henderson & Macquistan, 1993), yielding a sample of 3074 trials (exclusion of 3,94% of the trials).

All subsequent analysis concentrated on correct answers only, representing 97.59% of total trials. Our final dataset was composed of 3000 trials after exclusions of wrong answers.

6.2. Eye-tracking processing

Gaze-direction recorded from the headset was extracted by retrieving the yaw (i.e. horizontal rotation of the gaze vector, indicating how far the eyes are directed, left or right, relative to a forward facing reference) and pitch angles (i.e. vertical rotation, indicating how far the eyes are directed upward or downward) for each sample. Trials containing more than 50% missing eye-tracking data-points were excluded. To simplify analysis, we inverted the yaw angles when gaze was directed to the left side, so that all trials were represented as if the target appeared on the right (i.e. positive yaw values), allowing direct comparison between

conditions, irrespective of the side on which the target was presented. Median yaw was computed across the entire trial duration.

6.3. Data analysis

We used the R software version 2025.05.0+496 and Python version 3.12.4 from an anaconda environment. The scripts used for data analysis and figures, together with the data generated by the user responses and their eye movements, are available as supplementary material (S1).

Repeated-measure analysis of variance (ANOVA) was conducted on reaction times (RTs) and accuracy on all experimental conditions to test whether performance differed as a function of target spatial sector (front space and rear space), cue validity (valid and invalid), and distractor condition (zero, one near, one far and three). Mixed linear models were created with cue validity and distractor condition as fixed effects. Participant number was added as a random effect to control for baseline variability in individual response time. Results of the mixed linear model were completed with post-hoc pairwise comparisons.

Accuracy rates were obtained dividing the trials answered correctly by the total number of trials, then analysed according to cue validity and distractor condition, similar to reaction times.

To analyse eyetracking data, we followed the same method as reaction times and accuracy.

For eye-tracking data, trial-level median yaw values were submitted to non-parametric group comparisons (Mann-Whitney U) between cue validity, distractor conditions and spatial sectors, permitting us to compare the distribution of gaze and orientation of early gaze direction depending on experimental conditions.

We analysed the time to reach the area of interest (AOI), using eye movement recordings to calculate the time needed for gaze to reach the target zone. To do so, we first defined the AOI, which was computed as a sphere with a 20° radius around the center of the target, representing the actual size of the target object in the environment. The fixation cross also had a radius of 5°. The eye yaw location was extracted for each time point to determine when gaze left the fixation cross area and when it reached the target AOI. The dataset is composed of 2784 trials with a successful entry in the AOI. 188 trials did not reach the AOI, representing 6,75% of trials. We analysed the data obtained by conducting analysis of variances on linear mixed models with cue validity and distractor conditions as fixed effects.

An early gaze analysis was conducted. We observed the first 300 ms after cue onset to detect difference in behaviour between cue validity and distractor conditions to determine if the orientation of gaze differed in early time. To do so, the fixation cross was modelised using a 5° sphere, as the analysis conducted on time to AOI. The same statistical approach was used for this analysis.

Gaze onset latency, meaning the time that gaze took to leave the fixation cross area, was tested by computing gaze shift onset by extracting the gaze “departure” from the fixation cross area. For this analysis we used a radius of 15° to be able to observe more precisely differences in behaviour. This analysis is used to observe differences between front and rear spatial

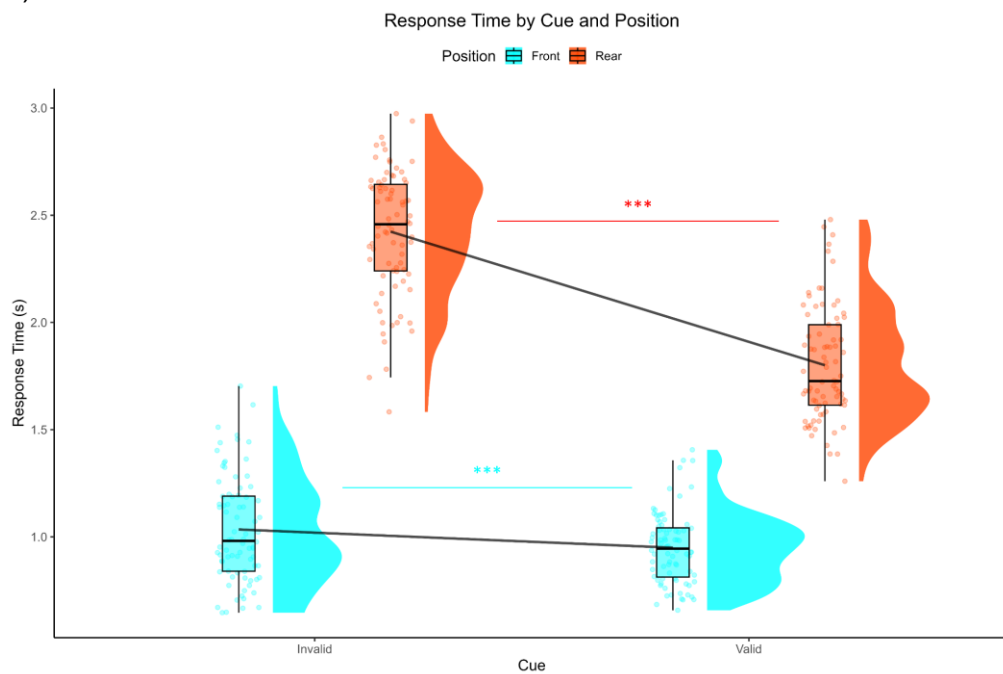
sectors of the target depending on cue validity. Instead of concentrating on the overall behaviour of early gaze, here we want to see if the direct orientation is impacted by our factors. Data was analysed following our latter procedures using linear mixed models and tested with ANOVA.

RESULTS

1. Response times

1.1. Main effects in full 3D space

a)



b)

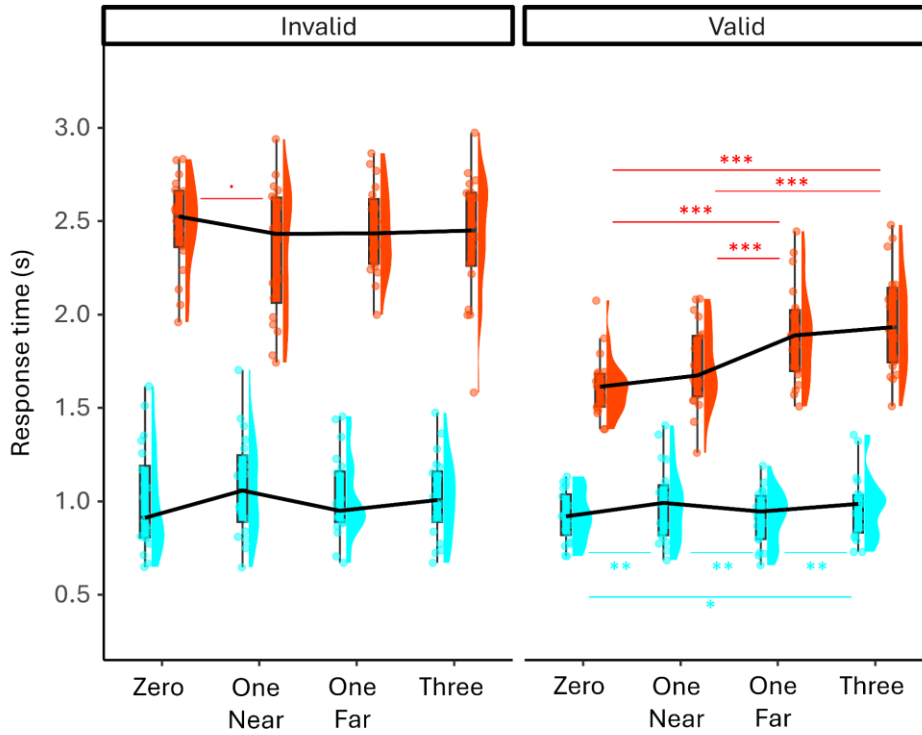


Figure 2: a) Response times depending on cue type and target position show main effects of cue validity (valid/invalid) and position (front/rear).

b) Response times depending on distractor condition and cue validity show a main effect of cue, with response times being slower in invalid trials, and a main effect of distractor. The more distractors present the slower are the responses.

For RTs, we observed a significant main effect of cue validity (figure 2a), $F(1, 19)=88.11$, $p<.001$, $\eta_p^2 = 0.82$, indicating that participants were generally faster for valid ($M = 1.38$, $SD = 0.48$) than for invalid ($M = 1.73$, $SD = 0.75$) trials, $t(299) = 5.037$, $p<.001$. We also found a main effect of distractor condition, $F(2.22, 42.09)=4.12$, $p<.001$, $\eta_p^2 = 0.18$, reflecting that participants were faster with no distractor ($M = 1.51$, $SD = 0.66$) than with one near ($M = 1.54$, $SD = 0.61$), one far ($M = 1.57$, $SD = 0.67$) $t(137) = 2.17$, $p=.032$, and three distractors ($M = 1.60$, $SD = 0.67$). There were also significant differences between one near and one far distractors, $t(137) = 1.98$, $p=.05$.

As expected, RTs also differed according to spatial sector, $F(1, 19)=1291.13$, $p<.001$, $\eta_p^2 = 0.99$, with generally faster detection of targets in front ($M = 0.99$, $SD = 0.21$) than rear space ($M = 2.11$, $SD = 0.42$), $t(299) = -31.99$, $p<.001$.

Finally, interactions between cue validity and distractor condition $F(3, 57)=7.30$, $p<.001$, $\eta_p^2 = .28$, cue validity and spatial sector $F(1, 19)=64.83$, $p<.001$, $\eta_p^2 = 0.77$, as well as the three-way interaction between cue validity, spatial sector, and distractor condition $F(3, 57)=5.56$, $p=.002$, $\eta_p^2 = 0.23$, were significant, suggesting that all three experimental factors could influence the effect of each other. This prompted us to decompose our analysis into different subsets of trial types, particularly to compare attentional modulations in rear and front space (see next section).

Similar results were obtained for accuracy (Table 2). This analysis revealed a main effect on cue validity, $F(1, 19) = 14.11$, $p<.001$, $\eta_p^2 = 0.43$, indicating that participants were less accurate

in invalid ($M = 0.89$, $SD = 0.18$) than valid trials ($M = 0.95$, $SD = 0.07$), $t(299) = 3.32$, $p=.001$; and a main effect of distractor condition $F(2.08, 39.55) = 4.97$, $p=.011$, $\eta_p^2 = 0.21$, reflecting that participant were more accurate with no distractor ($M = 0.94$, $SD = 0.14$) than with one distractor near ($M = 0.92$, $SD = 0.13$), one far ($M = 0.93$, $SD = 0.13$), or three ($M = 0.88$, $SD = 0.16$), although pairwise differences were not significant. A main effect of spatial sector was also found $F(1, 24)=36.66$, $p=.000$, $\eta_p^2 = 0.67$, indicating higher accuracy in front ($M = 0.98$, $SD = 0.06$) than rear space ($M = 0.80$, $SD = 0.25$), $t(299) = -23.37$, $p<.001$. Notably, both cue validity and distractor condition interacted with spatial sector, suggesting differences in performance for front and rear targets (table 2), but there was no significant three-way interaction. Therefore, further analysis of these data was also conducted by breaking down statistical comparisons for each spatial sector.

	<i>Sum sq</i>	<i>Error sq</i>	<i>DF num</i>	<i>Df den</i>	<i>F</i>	<i>p</i>	
<i>(Intercept)</i>	770.872	6.02	1	19	2434.41	>.000	***
<i>cue</i>	10.021	2.16	1	19	88.11	>.000	***
<i>position</i>	100.323	1.48	1	19	1291.13	>.000	***
<i>distractor</i>	0.377	1.74	3	57	4.12	.010	*
<i>cue x position</i>	5.785	1.70	1	19	64.83	>.000	***
<i>cue x distractor</i>	0.510	1.33	3	57	7.30	>.000	***
<i>position x distractor</i>	0.580	1.80	3	57	6.13	.001	**
<i>cue x position x distractor</i>	0.413	0.41	3	57	5.56	.002	**

Table 1. Type III Repeated-Measures ANOVA for all reaction times

	<i>Sum sq</i>	<i>Error sq</i>	<i>Df num</i>	<i>Df den</i>	<i>F</i>	<i>p</i>	
<i>(Intercept)</i>	270.113	0.597	1	19	8589.913	>.000	***
<i>cue</i>	0.313	0.421	1	19	14.107	.001	**
<i>position</i>	1.128	0.565	1	19	37.921	>.000	***
<i>distractor</i>	0.135	0.516	3	57	4.97	.003	**
<i>cue x position</i>	0.378	0.417	1	19	17.237	>.000	***
<i>cue x distractor</i>	0.039	0.620	3	57	1.193	.320	
<i>position x distractor</i>	0.093	0.477	3	57	3.714	.016	*
<i>cue x position x distractor</i>	0.028	0.565	3	57	0.955	.420	

Table 2. Type III Repeated-Measures ANOVA for Accuracy

1.2. Front space

When focussing on trials with targets appearing in front space only, results showed a main effect of cue validity $F(1, 133)=13.76$, $p<.001$, $\eta_p^2 = 0.09$, 95% CI [0.03, 1.00]), with RTs being 8% faster in valid trials ($M = 0.95$, $SD = 0.17$) than invalid trials ($M = 1.03$, $SD = 0.25$), $t(139) = -3.70$, $p<.001$. No significant effect of distractor condition was found $F(1, 133)= 2.09$, $p>.05$, nor any interaction between cue validity and distractor condition, $F(3, 133)=0.23$, $p=.874$.

However, closer inspection of the different trial types suggests that distractor effects reflected the visual perceptual similarity of conditions rather than their experimental configuration in space. Specifically, as evident in figure 1, distractors appearing in rear space were not immediately visible to the participant at their onset and unlikely to be seen when presented with a front target. We therefore recategorized these trials into distractor present/visible (one near or three) and distractor absent/invisible (zero or one far) conditions.

A new ANOVA on RTs after recategorization (Table 4) revealed not only a significant main effect of cue validity, $F(1, 137)=13.98, p<.001, \eta_p^2 = 0.09, 95\% \text{ CI } [0.03, 1.00]$, as above, but now also a significant main effect of distractor conditions, $F(1, 137)=4.89, p=.029, \eta_p^2 = 0.03, 95\% \text{ CI } [0.00, 1.00]$, with faster detection in the absent ($M = 0.97, SD = 0.21$) than the present ($M = 1.02, SD = 0.22$) conditions, $t(59) = -3.70, p < .001$. However, no interaction was observed between cue validity and distractor condition, $F(1, 133)=0.80, p=.494$, suggesting independent effects of spatial cueing and serial search.

ANOVAs conducted on accuracy rates (table 5, figure 3) showed no significant differences according to cue validity or distractor condition, neither with the 4 original nor the 2 recategorized conditions, presumably due to ceiling effects when targets were more easily reported in front space.

	<i>Sum sq</i>	<i>Mean sq</i>	<i>DF num</i>	<i>DF den</i>	<i>F</i>	<i>p</i>	
<i>cue</i>	0.289	0.289	1	133	13.76	>.000	***
<i>distractor</i>	0.132	0.044	3	133	2.094	.104	
<i>cue x distractor</i>	0.015	0.004	3	133	0.232	.874	

Table 3. Type II ANOVA for front reaction time

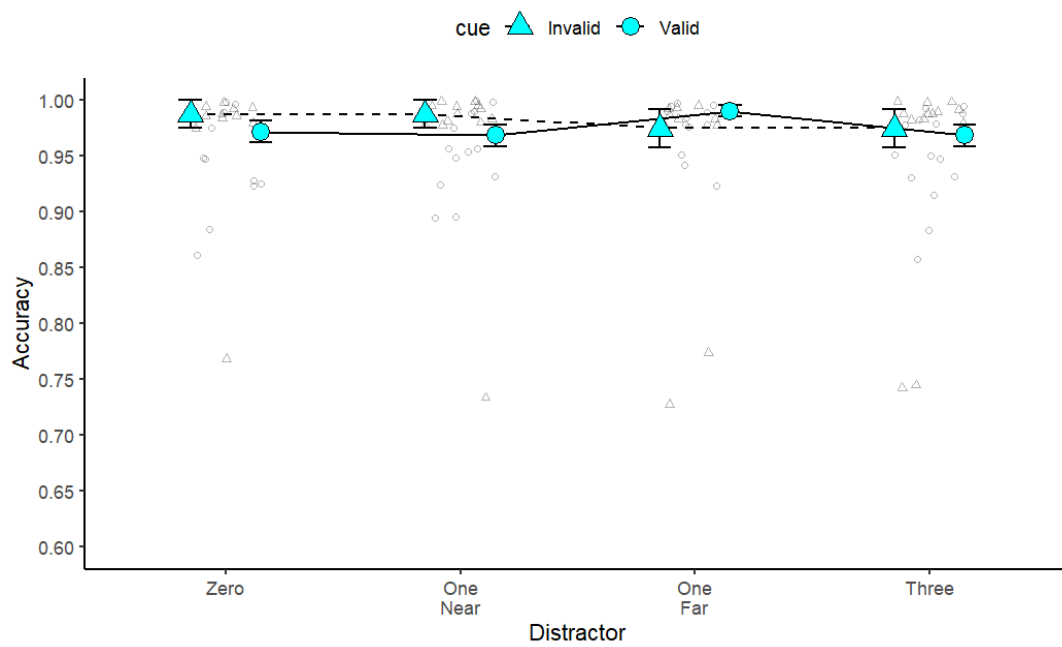
	<i>Sum sq</i>	<i>Mean sq</i>	<i>DF num</i>	<i>DF den</i>	<i>F</i>	<i>p</i>	
<i>cue</i>	0.289	0.289	1	137	13.976	>.000	***
<i>distractor</i>	0.101	0.101	1	137	4.886	.029	*
<i>cue x distractor</i>	0.007	0.007	1	137	0.317	.575	

Table 4. Type II ANOVA for front reaction time distractor recategorized

	<i>Sum sq</i>	<i>Mean sq</i>	<i>Df num</i>	<i>DF den</i>	<i>F</i>	<i>p</i>
<i>cue</i>	0.002	0.002	1	137	0.528	.469
<i>distractor</i>	0.002	0.002	1	137	0.528	.469
<i>cue x distractor</i>	0.002	0.002	1	137	0.528	.469

Table 5. Type II ANOVA for front recategorized accuracy

a)



b)

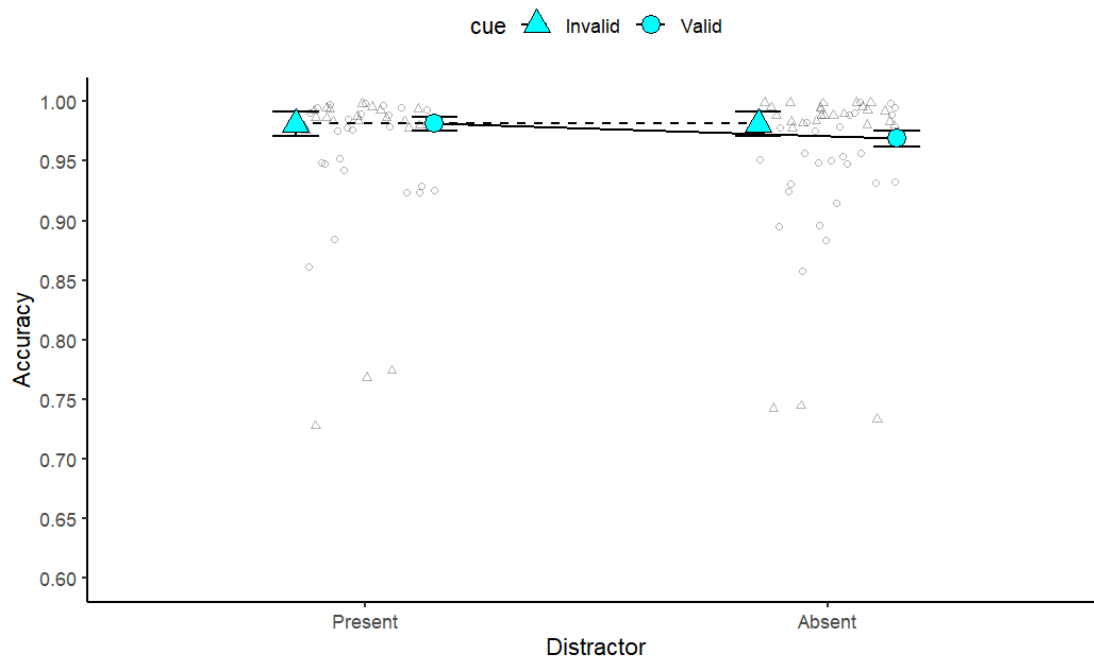


Figure 3: Accuracy in valid and invalid trials in front space before (a) and after (b) recategorization of the distractor types.

1.3. Rear space

A similar two-way repeated measure ANOVA was conducted on RTs for rear targets only (Table 6), equivalent to the analysis for front space above. This showed a main effect of cue validity, $F(1, 133)=281.28$, $p<.001$, $\eta_p^2 = 0.68$, 95% CI [0.61, 1.00]), with participants being 34% slower in invalid ($M = 2.42$, $SD = 0.29$) than valid trials ($M = 1.80$, $SD = 0.27$), $t(139) =$

15.42, $p < .001$. There was also a main effect of distractor condition, $F(1, 133)=4.99$, $p<.001$, $\eta_p^2 = 0.010$, 95% CI [0.02, 1.00]), and a significant interaction between cue validity and distractor condition, $F(3, 133) =5.49$, $p<.001$, $\eta_p^2= .11$, 95% CI [0.03, 1.00]).

As seen in figure 2b, this interaction was driven by the valid condition. Accordingly, after valid cues, RTs were significantly faster on trials with no distractor ($M = 2.05$, $SD = 0.48$) or one near distractor ($M = 2.03$, $SD = 0.43$), relative to those with one far ($M = 2.17$, $SD = 0.37$), $t(133) = -3.776$, $p < .0002$, or three distractors ($M = 2.19$, $SD = 0.38$), $t(133) = -4.714$, $p < .0001$. The significant difference between one near ($M = 0.81$, $SD = 0.21$) and one far distractor, $t(133) = -2.313$, $p < .0222$, indicates that participants are impacted by the position of the distractor in front space when the target appears in rear space. Because the near distractor is on the same side as the target (see figure 1), it is likely that orienting to the distractor will put it “on the way” toward the rear targets, whereas orienting to the far distractor will make participants turn toward the opposite direction, away from the target. As the latter condition will require reorienting to the target side, it will cause slower RTs and make the distractor effect similar to invalid spatial cueing. This also accords in turn with the significant *cue validity x distractor condition* interaction found in rear space.

Finally, the ANOVA on accuracy measures revealed a significant main effect of cue validity, $F(1, 133) =35.01$, $p<.001$, $\eta_p^2= .21$, 95% CI [0.11, 1.00]), with lower detection in invalid ($M = 0.79$, $SD = 0.21$) than valid trials ($M = 0.92$, $SD = 0.09$), $t(139) = -5.74$, $p < .001$; and a significant main effect of distractor condition, $F(1, 133) =3.82$, $p<.05$, $\eta_p^2= .08$, 95% CI [0.01, 1.00]). Accordingly, accuracy tends to decrease with the number of distractors. Participants perform worse when three distractors ($M = 0.8$, $SD = 0.17$) are present compared to one far ($M = 0.88$, $SD = 0.17$) $t(137) = 2.23$, $p < .05$, one near ($M = 0.87$, $SD = 0.16$) $t(137) = 2.05$, $p < .05$, or when no distractor are present ($M = 0.90$, $SD = 0.18$) $t(137) = 2.85$, $p < .01$. (cf. table 7).

	<i>Sum sq</i>	<i>Mean sq</i>	<i>Df num</i>	<i>DF den</i>	<i>F</i>	<i>p</i>	
<i>cue</i>	15.517	15.516	1	133	281.284	>.000	***
<i>distractor</i>	0.826	0.275	3	133	4.989	.002	**
<i>cue x distractor</i>	0.909	0.302	3	133	5.492	.001	**

Table 6. Type II ANOVA for rear reaction time

	<i>Sum sq</i>	<i>Mean sq</i>	<i>Df num</i>	<i>DF den</i>	<i>F</i>	<i>p</i>	
<i>cue</i>	15.517	15.517	1	133	281.284	>.000	***
<i>distractor</i>	0.826	0.275	3	133	4.989	.003	**
<i>cue x distractor</i>	0.909	0.301	3	133	5.492	.001	**

Table 7. Type II ANOVA for rear accuracy

2. Eye-tracking results

2.1. Gaze distribution in space

To further examine how participants oriented attention in the 360° spatial surroundings as a function of exogenous cueing and endogenous search demands, we also exploited continuous eye-tracking data recorded from the VR headset. Heatmaps of gaze density across trials were obtained by pointing each time point of gaze in the environment along the yaw and pitch axes of space around the participant, during each trial (from cue onset to target detection) and for each experimental condition (fig. 6).

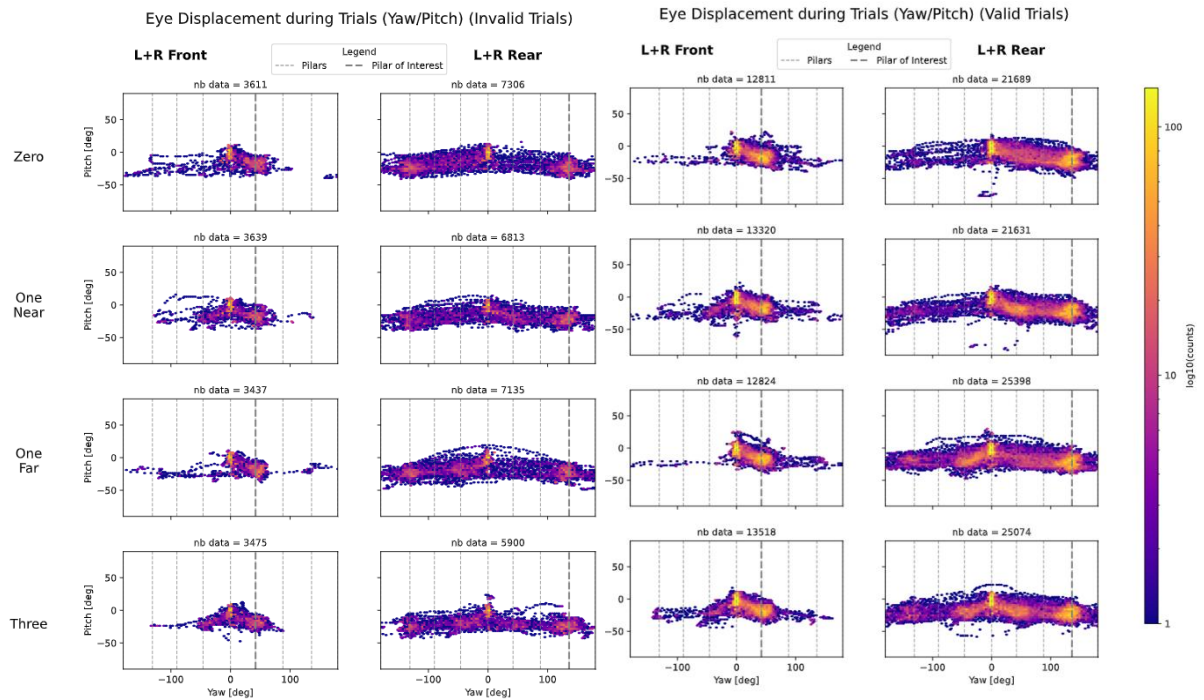


Figure 4: Heatmap of eye displacement representing the position of gaze (yaw in degree) in the environment during invalid (2 left columns) and valid (2 right columns) conditions depending on distractor type : zero (first line), one near (second line), one far (third line), and three (fourth line). The x axis represents both front space (from starting position at 0deg to 90deg) and rear space (from 90deg to 180deg). Each pillar position is indicated by thin dotted lines, and target position by thick dotted lines.

The heatmaps of gaze position (figure 4) show a clear difference between front and rear space in terms of density and distribution, as well as clear effects of cue validity and distractor condition. During valid trials, gaze was mainly directed to the target, with a path clearly apparent between two hot spots corresponding to the initial fixation cross and the target, in both front and rear space. The presence of one (visible) or more distractors induced more frequent gaze toward their location or the other side, particularly when the target appeared in rear space. Invalid trials show broader visual exploration, with clear deviation opposite to the target side and attenuated gaze path density to the target even without distractors, confirming an effective impact of invalid cues. There was also a reduced density of fixations on the targets and distractors compared to valid trials. More quantitative analysis of these gaze data are reported in next sections.

2.1.1. Cue validity

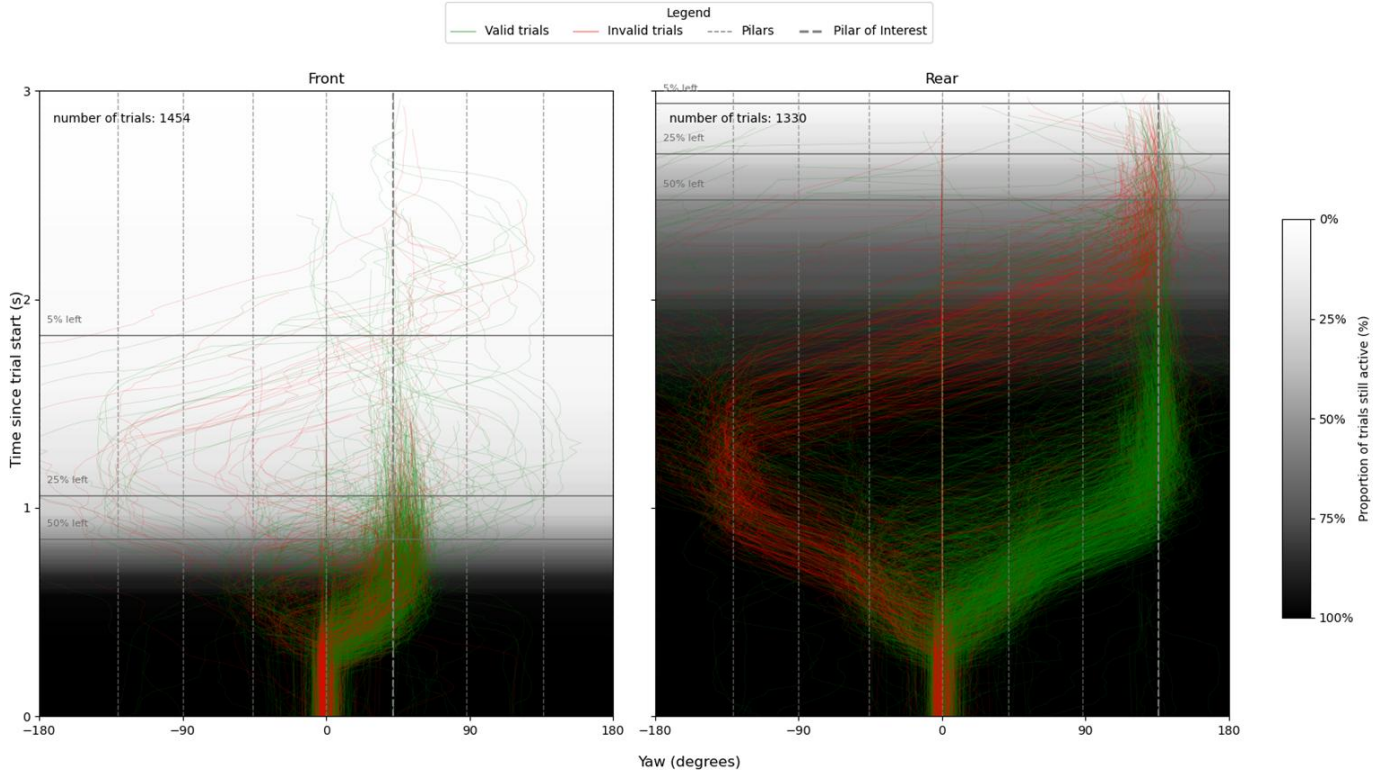


Figure 5: Horizontal eye rotation during trials. The red lines represent the gaze of participants during invalid trials, the green lines represent the gaze during valid trials. The colour scale from black to white represents the timespan of trials. The central line represents the fixation cross position. The dotted lines at 45° and 135° represent the location of the target for front and rear space, respectively.

Since most eye movements occurred on the yaw plane, our subsequent analysis considered only eye positions along this axis. Validity effects were best evidenced by plotting gaze displacement in yaw over the whole trial duration, separately for valid and invalid trials as shown in figure 5. This demonstrated clear deviations to the incorrect side (red traces) followed by returning to the correct side (green traces) on invalid compared to valid trials, for both front and rear target conditions. Analysis of eye positions across all timepoints within a trial confirmed strong validity effects. For front space, a Mann-Whitney U test revealed highly significant differences in yaw positions between valid and invalid trials, with a sustained lateralized gaze bias in the former ($M = 19.74$, $SD = 29.98$) significantly larger than in the latter case ($M = 12.99$, $SD = 35.82$), $U = 4.07e8$, $n_1 = 52473$, $n_2 = 14162$, $z = 17.5$, $r = .07$, $p < .001$. Similar results were observed for rear space: valid trials produced a strong and sustained lateralized gaze shift ($M = 47.72$, $SD = 75.83$), whereas invalid trials presented a significantly smaller median angle with initial gaze shift opposite to the target ($M = 5.84$, $SD = 94.38$), $U = 1.61e9$, $n_1 = 93792$, $n_2 = 27154$, $z = 67.23$, $r = .19$, $p < .001$.

2.1.2. Distractor effect

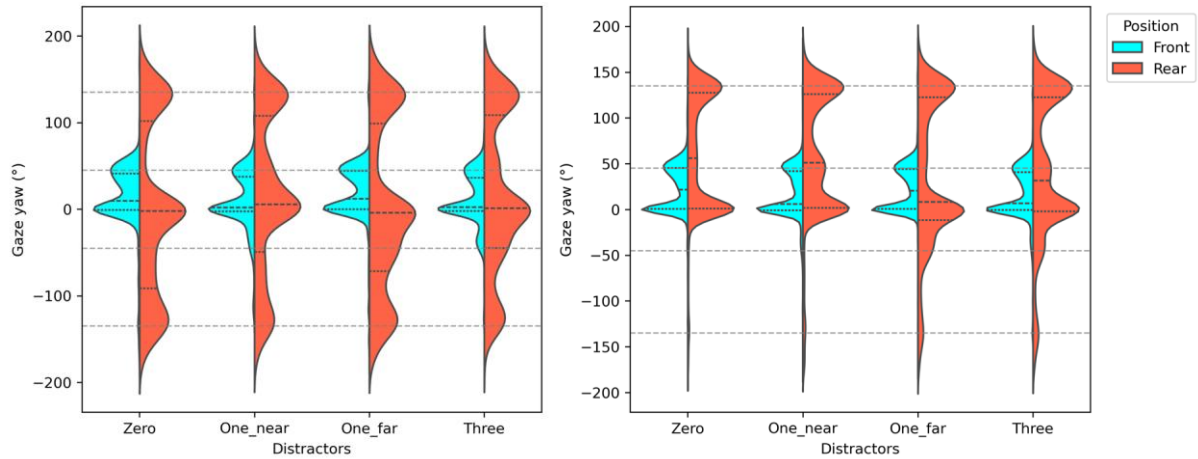


Figure 6: Gaze angle distribution depending on distractor on invalid (left) and valid (right) trials. Distribution plots indicate the number of gaze samples at each location along the horizontal yaw axis, across all trials from each condition. The fixation cross is located at 0° angle, targets appear at 45° (front) or 135° (rear), while distractors appear at either of these locations or intermediate locations at -45° and -135°. Dash lines correspond to the positions of the pillars in space.

To analyse the effect of distractor conditions, we then concentrated on valid trials only. Distractor effects were quantified by computing gaze distribution over the yaw axis across the different conditions, as plotted in figure 6 (additional details presented in supplementary material tables). Statistical analysis confirmed differences in gaze distribution according to both distractor condition and spatial sector. In front space, trials without distractors produced the highest gaze focus on the target ($M = 21.05$, $SD=31.61$), relative to trials with one distractor near ($M = 15.11$, $SD=32.19$), $U=1.56e8$, $n1 = 16422$, $n2=16959$, $z= 19.36$, $r=.11$, $p<.001$; or trials with three distractors ($M = 14.93$, $SD=31.54$) $U=1.56e8$, $n1 = 16422$, $n2=16993$, $z= 19.18$, $r=.10$, $p<.001$ (table 8).

Trials with one distractor far (i.e., not visible) showed similar difference ($M = 22.39$, $SD=29.53$) in comparison to one distractor near, $U=1.19e8$, $n1 = 16959$, $n2=16261$, $z= -21.54$, $r=.12$, $p<.001$; or three distractors, $U=1.57e8$, $n1 = 16261$, $n2=16993$, $z= 21.32$, $r=.12$, $p<.001$; while there was no significant differences between zero distractor and one distractor far, $U=1.32e8$, $n1 = 16422$, $n2=16261$, $z= -1.76$, $r=.01$, $p=.08$, or between three distractors and one distractor near, $U=1.44e8$, $n1 = 16993$, $n2=16959$, $z= -0.38$, $r=.00$, $p=.70$.

The analysis for rear space (table 9) replicated these effects even more clearly, with significant differences found between all distractor conditions. Gaze distribution over the target in the zero distractor condition ($M = 44.13$, $SD=80.78$) differed from one distractor near ($M = 46.77$, $SD=77.83$) $U=4e8$, $n1 = 28995$, $n2=28444$, $z= -6.28$, $r=.03$, $p<.001$, as well as from one far ($M = 28.46$, $SD=86.39$) $U=5.34e8$, $n1 = 28995$, $n2=32533$, $z= 28.51$, $r=.11$, $p<.001$, and three ($M = 35.47$, $SD=81.82$) $U=4.79e8$, $n1 = 28995$, $n2=30974$, $z= 14.09$, $r=.06$, $p<.001$. These differences demonstrate that participants oriented their gaze less effectively to the target when one or more distractors were present. There was also a significant difference between one near and one far distractors, $U=5.35e8$, $n1 = 28444$, $n2=32533$, $z= 33.41$, $r=.14$, $p<.001$, and between one near and three distractors, $U=4.83e8$, $n1 = 28444$, $n2=30974$, $z= 20.16$, $r=.08$,

$p < .001$, presumably reflecting that a unique near distractor appears “on the way” of shifting toward the target, whereas distractors at other locations appear further away from the target and induce initial shifts in attention towards the opposite side. Similarly, one distractor far elicited a weaker distribution of gaze toward the target than three distractors, $U=4.71e8$, $n1 = 32533$, $n2=30974$, $z = -14.05$, $r = .06$, $p < .001$, indicating a greater probability to orient away from the target, i.e., at the diagonally opposite location, with a unique distractor far than three distractors spread in space.

Variable	Valid_Zero median	Valid_One_near median	n1	n2	U	z	p	r	Valid_One_far median	Valid_Three median
Valid_Zero vs Valid_One_near	21.819113	6.161261	12811	13320	96074867.0	17.64	0.0	0.11		
Valid_Zero vs Valid_One_far	21.819113		12811	12824	82597810.0	0.77	0.444	0.0	20.462032	
Valid_Zero vs Valid_Three	21.819113		12811	13518	97679060.0	17.99	0.0	0.11		7.044322
Valid_One_near vs Valid_One_far		6.161261	13320	12824	74899092.0	-17.23	0.0	0.11	20.462032	
Valid_One_near vs Valid_Three		6.161261	13320	13518	90157068.0	0.2	0.841	0.0		7.044322
Valid_One_far vs Valid_Three			12824	13518	97501378.0	17.55	0.0	0.11	20.462032	7.044322

Table 8. Front - Mann-Whitney comparisons within valid conditions

Variable	Valid_Zero median	Valid_One_near median	n1	n2	U	z	p	r	Valid_One_far median	Valid_Three median
Valid_Zero vs Valid_One_near	56.317312	51.319632	21689	21631	235684625.0	0.85	0.395	0.0		
Valid_Zero vs Valid_One_far	56.317312		21689	25398	328175251.0	35.88	0.0	0.17	8.302169	
Valid_Zero vs Valid_Three	56.317312		21689	25074	309270059.0	25.66	0.0	0.12		31.739244
Valid_One_near vs Valid_One_far		51.319632	21631	25398	322728490.0	32.74	0.0	0.15	8.302169	
Valid_One_near vs Valid_Three		51.319632	21631	25074	306003266.0	23.96	0.0	0.11		31.739244
Valid_One_far vs Valid_Three			25398	25074	303062360.0	-9.38	0.0	0.04	8.302169	31.739244

Table 9. Rear - Mann-Whitney comparisons within valid conditions

2.2. Time to reach AOI

2.2.1. Cue validity

We next analyzed the temporal latencies of gaze shifts from fixation cross to first fixation on the target. Results again showed reliable main effects of cue validity $F(1, 2564.4)=437.67$, $p < .001$, $\eta_p^2 = 0.16$, 95% CI [0.14, 1.00]) and spatial sector $F(1, 2567)=4666.12$, $p < .001$, $\eta_p^2 = 0.61$, 95% CI [0.60, 1.00]).

When considering front space only, the main effect of cue validity was significant, $F(1, 1305)=20.27$, $p<.001$, $\eta_p^2 = 0.68$, 95% CI [0.61, 1.00]), with faster fixation on the target AOI in valid ($M= 0.49$, $SD= 0.21$) than invalid trials ($M= 0.56$, $SD= 0.31$) $t(1305)= 4.504$, $p<.001$. The main effect of cue validity was also significant for rear targets, $F(1, 1244.1)=506.52$, $p<.001$, $\eta_p^2 = 0.28$, 95% CI [0.25, 1.00]), with faster gaze to the AOI in valid ($M= 1.22$, $SD= 0.41$) than invalid trials ($M= 1.81$, $SD= 0.45$) $t(1244)= 22.03$, $p<.001$.

2.2.2. Distractor effect

Gaze shift latencies also showed a main effect of distractor condition $F(3, 2562.7)=34.70$, $p<.001$, $\eta_p^2 = 8.47\text{e-}03$, 95% CI [0.00, 1.00]). This effect was present for targets in front space $F(3,1304.1)=21.11$, $p<.001$, $\eta_p^2 = 0.03$, 95% CI [0.02, 1.00]), with faster shifts to reach the AOI with no distractor ($M=0.46$, $SD=0.25$) than with one near ($M=0.56$, $SD=0.26$) $t(1304)= -4.832$, $p<.001$, or three distractors ($M=0.55$, $SD=0.25$) $t(1304)= 4.585$, $p<.001$; and likewise faster shifts with one distractor far (not visible) ($M=0.46$, $SD=0.18$) than with one near, $t(1304)= -4.393$, $p<.001$. This accords with the lack of immediate visibility of the far distractor at target onset in front space, accounting for the lack of differences in gaze shift latencies between one near and no distractor $t(1304)= 0.416$, $p=.678$.

The effect of distractors was also present for targets in rear space $F(3,1242.1)=34.94$, $p<.001$, $\eta_p^2 = 0.02$, 95% CI [0.01, 1.00]), where gaze shifts were again faster with no distractor ($M=1.21$, $SD=0.44$) than one far ($M=1.43$, $SD=0.46$) or three ($M=1.44$, $SD=0.48$). In addition, in rear space, gaze shifts were faster with one distractor near ($M=1.25$, $SD=0.49$) than one far $t(1242)= -4.288$, $p<.001$. In contrast, there was no significant difference between no distractor and one distractor near $t(1242)= 0.974$, $p=.330$, or between one distractor far and three $t(1242)= 0.904$, $p=.366$.

Overall, these gaze latency results follow the general pattern of reaction times as described earlier (section 1, RTs). Accordingly, we computed a regression between RTs and gaze latencies toward the target, across all trials but keeping each trial labeled for the amount of distractors concurrently present. This analysis revealed a strong correlation between RTs and gaze in both front space $r=.94$, and rear space $r=.79$, aligned with a globally increasing cost of increasing distractors (Figure 7).

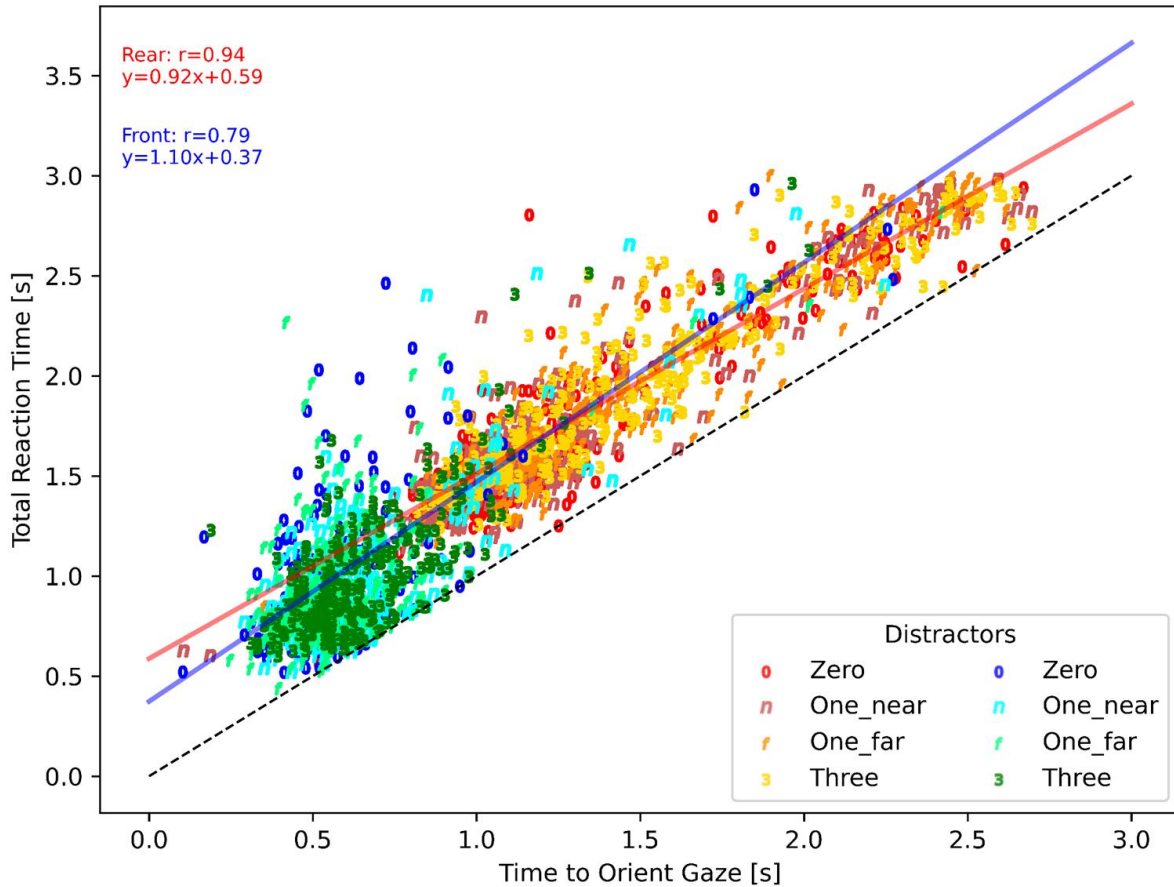


Figure 7 : Regression analysis showing the correspondence between gaze shift latencies and reaction time.

2.3. Early gaze movements

Because time to reach the target AOI reflected the total time needed for voluntary exploration of space, primarily driven by endogenous attention and constrained by physical gaze-head turns, we performed an additional analysis focusing on the initial part of trials and considering gaze position only within the first 300 ms timeframe after cue onset, potentially more sensitive to more rapid exogenous processes.

An ANOVA revealed significant effects of spatial sector $F(1, 41723)=7.91, p=.005, \eta_p^2 = 3.94e-04$, 95% CI [0.00, 1.00][5]) distractor condition, $F(1, 41722)=9.07, p<.001, \eta_p^2 = 5.39e-04$, 95% CI [0.00, 1.00]). No significant effect of cue validity was found, $F(1, 41727)=0.16, p = .69, \eta_p^2 = 1.15e-05$, 95% CI [0.00, 1.00]). However there were significant interactions between cue validity and spatial sector $F(1, 41722)=8.91, p=.003, \eta_p^2 = 2.16e-04$, 95% CI [0.00, 1.00]), and between distractor condition and spatial sector $F(3, 41721)=11.74, p<.001, \eta_p^2 = 2.05e-04$, 95% CI [0.00, 1.00]).

When separating each spatial sector (front / rear), the validity effect was marginal in front space, $F(1, 21783)=3.29, p=.069, \eta_p^2 = 1.51e-04$, 95% CI [0.00, 1.00]), but robust in rear space $F(3, 19932)=5.41, p=.02, \eta_p^2 = 2.76e-04$, 95% CI [0.00, 1.00]). In the latter case, early gaze movements were significantly quicker in the direction of the target on valid ($M=0.47, SD=6.05$) than invalid trials ($M=-0.12, SD= 14.62$) $p= .019$.

Likewise, the effect of distractor condition was not significant in front space, $F(3, 21783)=1.93$, $p=.12$, $\eta_p^2 = 1.51\text{e-}04$, 95% CI [0.00, 1.00]), but significant in rear space $F(3, 19925)=16.88$, $p<.001$, $\eta_p^2 = 1.42\text{e-}03$, 95% CI [0.00, 1.00]). Pairwise comparisons revealed that early gaze was oriented faster toward the target location with one distractor near ($M=1.30$, $SD=0.47$) than one distractor far ($M=-0.10$, $SD=0.47$), $p<.001$, or three distractors ($M=-0.41$, $SD=0.48$) $p<.001$. This adds to previous findings that one distractor near had a relative advantage due to orienting participants in the direction of a target appearing behind them.

2.4. Gaze onset latency

Finally, to further evaluate rapid and possibly reflexive orienting, we computed gaze shift onset by extracting the eye “departure” time from the initial fixation cross, i.e., when gaze left the central visual area of fixation (defined by a 15° radius around fixation cross as above).

This analysis revealed significant main effects of spatial sector $F(1,2694.3)=77.95$, $p<.001$, $\eta_p^2 = 0.01$, 95% CI [0.01, 1.00]) and distractor condition $F(3,2694.1)=11.92$, $p<.001$, $\eta_p^2 = 8.90\text{e-}03$, 95% CI [0.00, 1.00]), combined with an interaction between these two factors $F(3,2694.1)=32.33$, $p<.001$, $\eta_p^2 = 0.03$, 95% CI [0.02, 1.00]). This indicates that participants were slower to initiate attention/gaze shifts when the target appeared in rear space ($M=400.45$, $SD=192.52$) compared to front ($M=355.42$, $SD=159.24$), and more so in the presence of distractors. No significant cue validity effects were found, $F(1,2694.2)=0.068$, $p=.794$, indicating that attention/gaze shifts onset were not impacted by the current validity probability manipulation.

When separating each spatial sector, we observed a significant effect of distractor condition in both front $F(3,1372.1)=3.53$, $p<.01$, $\eta_p^2 = 6.44\text{e-}03$, 95% CI [0.00, 1.00]) and rear $F(3,1304.2)=37.09$, $p<.001$, $\eta_p^2 = 0.06$, 95% CI [0.04, 1.00]). However, the gaze onset pattern differed in rear space compared to front space as a function of distractor condition.

For targets in front space, gaze onset was significantly faster when no distractor was present ($M = 337$, $SE = 24.1$) compared to all other conditions with distractors, either one far ($M = 363$, $SE = 24.1$), $t(1372) = 2.14$, $p = .033$, one near ($M = 368$, $SE = 24.1$), $t(1372) = 2.61$, $p = .009$, and three ($M = 367$, $SE = 24.1$), $t(1372) = 2.47$, $p = .014$. No significant differences were found between one far and one near, $t(1372) = -0.45$, $p = .653$, between one far and three, $t(1372) = -0.31$, $p = .757$, or between one near and three, $t(1372) = 0.14$, $p = .887$.

In contrast, for targets in rear space, gaze onset was significantly longer when no distractor was present ($M = 478$, $SE = 29.4$) compared to one far ($M = 370$, $SE = 29.4$), $t(1304) = -7.52$, $p < .001$, one near ($M = 376$, $SE = 29.5$), $t(1304) = -7.03$, $p < .001$, and three distractors ($M = 380$, $SE = 29.7$), $t(1304) = -6.52$, $p < .001$. However, there was no significant difference between one far and one near distractors, $t(1304) = -0.46$, $p = .646$, between one far and three distractor, $t(1304) = -0.70$, $p = .486$, or between one near and three distractors, $t(1304) = -0.26$, $p = .799$. This pattern of behaviour in gaze onset latency might reflect a non-spatial alerting effect of visual stimuli appearing in front of the participant when the sound is placed behind them, possibly explaining why the presence of a distractor could facilitate gaze shift onset for the rear condition but not the front (unlike their effect on shifting direction and latency).

DISCUSSION

1. Attentional orienting and cueing effects

Our study reveals that spatial attention processes can reliably be tracked in a full 360° space through behavioral response indices and ongoing eye-tracking measures using a novel immersive VR paradigm, paving the way to more ecological approaches of attention in 3D. Consistent with our hypotheses, valid auditory cues facilitated target detection while invalid cues slowed it, although cue location had no predictive value. This effect emerged in both front and rear space. Reaction times were faster when cues correctly indicated the target location and slower when they misdirected attention, regardless of whether the target appeared in front or behind the participant. This demonstrates that exogenous spatial shifts of attention occur in rear space through mechanisms similar to those governing attention in front space. These results complement existing literature and highlight the fact that multisensory information is integrated in shared spatial representations to guide attention. Here we show that this also occurs when attention shifts to locations outside the participant's direct line of sight.

Converging evidence for this cueing effect was found in eye-tracking data. The analysis of latencies to reach the target Area of Interest (AOI) and gaze dispersion revealed that participants were faster to look to the target when the sound cue was valid, regardless of target location. This was also apparent in the gaze path found on valid trials, which were more focused, as opposed to more dispersed visual exploration on invalid trials. Furthermore, the early gaze path trajectories clearly showed that participants quickly oriented their eyes toward the sound source location and explored space opposite to the target after invalid cues, suggesting that an exogenous attention capture (automatic, bottom-up attention shifts triggered by a salient stimulus) occurred in rear space and thus delayed the detection of a subsequent target in front space. These results are consistent with classic Posner cueing effects (Chica, Bartolomeo & Lupiáñez, 2013; Kahneman & Henik, 1981/2017; Posner, Cohen & Rafal, 1982) and extend previous work typically limited to 2D screens.

Although previous research already examined attention to events behind participants, these studies remain scarce. Moreover, most were restricted to visual targets seen through mirrors in the front space, and did not require participants to actually turn in physical space when directing attention to objects behind them in a naturalistic setting. Based on these results, (Spence, Lee & van Der Stoep, 2020) suggested that spatial representations governing attention in rear space might differ fundamentally from front space. Our findings point to a different interpretation. The difference may arise because the mirrors used in (Spence et al., 2020) and Soret et al., (2022) required a mental rotation to detect a target, whereas in our experiment participants could see directly the target by turning their eyes and body in real 3D space. The slower responses found in these studies for targets in the rear might be explained by the mental transformation involved in processing targets seen in the mirror. In this interpretation, the slower responses would not result from attentional mechanisms, but rather reflect additional cognitive processing due to the mental transformation. Here we found that auditory cues trigger spatial attention shifts to unseen rear locations, such that invalid cues misdirect attention and bias initial gaze movements away from the target location. These results suggest that attention orienting relied on similar mechanisms for both front and back

space, with a general difference in response times driven mainly by motor demands inherent to moving the eyes and body to locations behind the participant due to the naturalistic 3D setting.

The dominance of visual information in guiding attention orienting is also relevant to explain our results. At trial onset, participants had a direct visual perception of the front space scene and subsequent stimuli (target or distractor), reinforced or attenuated by the auditory cue. In contrast, rear space perception initially depended only on auditory cues, with relevant visual information perceived only when participants turned. In addition, we observed striking differences in the early gaze-shift onset (i.e., when the eyes first moved toward the target or another location), depending on the number of distractors. Thus, with fewer distractors, early gaze onset was slower, but response time faster. This suggests that irrelevant visual events could facilitate attention mobilization, presumably via non-spatial alerting mechanisms, while the same visual cues could then slow attention shifts and gaze, and therefore produce longer time to reach the target, a pattern further supporting an important visual dominance in attention orienting across space.

This observation accord more generally with our conclusion that front and rear spaces are not linked to qualitatively different attention systems, but that potential differences between front and rear regions might emerge from the need to move physically in space to see targets behind oneself and the inherent visual dominance of perceptual processes governing attention shifts in our multimodal 3D setting.

2. Serial search

Results on serial search effects in our VR paradigm are coherent with previous work: adding distractors in the environment slows target detection. Distractors matched the target in colour and several shape features, increasing task difficulty in accordance with target-distractor similarity principles (Olk et al., 2018; Poiese et al., 2008; Treisman, 1991). Furthermore, increasing the number of distractors requires attention to shift serially across multiple locations for target discrimination, slowing detection times in direct proportion to the amount of distractors presented.

In front space, we effectively only tested the presence of one distractor in the visual field, and it did significantly delay the participant's responses. Eye-tracking data also revealed that gaze initially deviated towards the distractor even though it was not beneficial to the task, thus confirming that distractors capture attention.

However, for rear space, where the task involved a coordinated visual and motor response (i.e., the participant had to turn), more nuanced results were observed. First, and critically, serial search effects did emerge when orienting attention to the rear, with generally slower reaction times when more distractors were presented. However, these effects depended not only on the number of distractors but also on their location. In particular, when only one distractor was present, its impact was determined by its relative distance to the target. A single distractor directly visible in front space and on the same side/ same hemispace than the rear target did not significantly increase the time needed to find the target. Nevertheless, eye-tracking analysis revealed a consistent increase of gaze position density over the distractor near position in addition to the final target location (see Figs. 5 and 7), not seen in the absence

of distractor. This pattern suggests a double influence of this distractor condition as its position was “on the way” toward the target to be reported behind the participant. Thus, while the near distractor probably captured attention to some degree (as evidenced in eyetracking), this initial involuntary shift arose in the correct hemifield and indirectly cued attention to further shift in the same direction toward the rear target, without any significant cost on detection response times. On the contrary, a single distractor placed on the opposite side/hemisphere relative to the rear target (far condition) produced a strong diversion of attention and incorrect gaze shifts to the non-relevant side, leading to subsequent reorienting and significantly detection responses. This interpretation is also consistent with the absence of significant difference between a distractor placed far (condition one far) and the presence of three distractors, when targets were presented behind the participant. In both cases, the distractor seen at the far position in front space is likely to initiate an attention shift towards the wrong hemifield, inducing slower responses despite any indirect cueing at the near distractor location “on the way” to the target. More generally, our results show that eye-tracking data offer a sensitive measure of subtle and transient changes in attention focus over time during a naturalistic task, even in the absence of concomitant behavioral effects in response times.

This interpretation is further confirmed by the significant interaction between cue validity and distractor condition in rear space found for RTs. When an invalid cue misdirected attention to the wrong hemifield and distractors were simultaneously present, target detection times increased, due to additional search in the environment. Participants first oriented in the wrong direction, then presumably searched within that space, before correcting their behaviour and executing a visuomotor reorientation toward the other hemifield, with transient attentional engagement and disengagement through multiple distractors to search for and eventually identify the target. This did not happen in front space, due to the nature of our experimental design: there, the target appears in visible space, and the effect of visual dominance discussed above will dominate on early search behavior.

Overall, these results support the interpretation that serial search operates in the rear space in a similar fashion as it does in front space. This accords with another study (Plewan & Rinkenauer, 2021) demonstrating that multiple distractors may impair detection performance when placed in a circular manner in depth around a target, presented in front space. Our results reveal that this serial search effect is not limited to 2D or front-view tasks, and that it generalizes to more ecological 3D environments with a full 360° space extending behind the participant. However, they also nuance the serial search effect typically caused by task-irrelevant distractors: when visuomotor integration is relevant for the task, as it is when the target is presented in rear space (at initially unseen location), the position of distractors may influence attentional shifts and exploration in a partly similar way as a spatial cue, particularly when the distractors appear directly at a visible location before the target is visible, and/or when the distractor location is along the visual exploratory trajectory towards the target. In these naturalistic conditions, distractor location in space may become as important as the number of distractors present and thus modulate (or even abolish) apparent attention capture effects observed in search.

CONCLUSION

Our VR-based approach provides a novel and sensitive tool to study attention deployment in both front and rear space at the same time, integrating both visual and auditory stimuli in naturalistic conditions. Importantly, the presence and number of distractors in front of as well as behind the participant, as well as the influence of valid or invalid spatial cues and their interaction, suggests a continuity in space representation governing attention shifts to seen and unseen locations in ecological environments. Remarkably, eye tracking data acquired in our VR setting offered a robust parallel to behavioral effects and successfully tracked gaze pattern differences that neatly reflected attentional orienting in both space and time, in ways not always captured by behavioral measures (e.g. RT) only.

Specifically, our results demonstrate a consistent effect of auditory spatial cues on attention shifts. These effects operate similarly in front and rear spaces, supporting at least partly shared mechanisms for sensory integration and exogenous attention shifts both in front of and behind the participant. The differences found between these sectors of personal space can be explained by visual dominance over auditory stimuli.

We also found evidence for serial search effects caused by concomitant distractors in both front and rear space, paralleling those reported in standard experiments conducted in front space. However, the sensorimotor integration inherent in having to physically turn during search in real 3D space points to more complex and nuanced view on typical serial search patterns, where the location of distractors may become as important as their number. However, many questions on the nature of space representation in attention deployment still stands. Further research is needed to confirm whether attention deployment operates similarly in front of the person, or behind them. For example, it remains to determine whether space perception in terms of allocentric or egocentric reference frames impacts attentional deployment differently for stimuli in front of or behind the person. Another question concerns whether a single continuous spatial priority map (**REF) is generated for front and back to represent target and distractor information when parts of task-relevant space is not visible. The relationship between different sensory modalities in these mechanisms will also require further investigation. In any case, our study encourages future research to use immersive and dynamic 3D environments for further characterizing selective attention in space.

SUPPLEMENTARY MATERIALS

S1. Anonymized user data and scripts for data analysis and generation of the figures, mannwhitney tables for gaze dispersion.

S2. Video showing the experimental procedure and summarizing the same results.

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